12TH CHEMISTRY · CHAPTER 17 · SOLVED EXERCISE

Group 2 Elements

Complete textbook exercise (MCQs, Short Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS

I) Which Group 2 oxide forms the most alkaline solution in water?

(a) Beryllium oxide
(b) Magnesium oxide
(c) Calcium oxide
(d) Strontium oxide

Answer: (d) Strontium oxide — The basic character of the oxides increases down the group, and strontium is the lowest member among the options.

II) Which Group 2 element does not react with water or steam?

(a) Beryllium (Be)
(b) Magnesium (Mg)
(c) Calcium (Ca)
(d) All Group 2 metals react

Answer: (a) Beryllium (Be) — Beryllium does not react even when heated, because of its protective oxide layer.

III) What are the products when magnesium reacts with steam?

(a) $Mg(OH)_2$ and H_2
(b) MgO and H_2
(c) $Mg(OH)_2$ only
(d) MgO and H_2O

Answer: (b) MgO and H_2 — $Mg_{(s)} + H_2O_{(g)} \rightarrow MgO_{(s)} + H_{2(g)}$. The hydroxide forms only with cold water.

IV) The species formed when a Group 2 basic oxide (e.g., CaO) reacts with water:

(a) A salt and hydrogen
(b) A metal hydroxide
(c) A metal carbonate
(d) A salt and water

Answer: (b) A metal hydroxide — The oxide ion reacts with water to give OH^- ions: $CaO + H_2O \rightarrow Ca(OH)_2$.

V) Which Group 2 carbonate is the most thermally stable?

(a) $BeCO_3$
(b) $MgCO_3$
(c) $CaCO_3$
(d) $BaCO_3$

Answer: (d) $BaCO_3$ — Thermal stability increases down the group; $BaCO_3$ decomposes at the highest temperature (1360 °C).

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VI) What are the products of heating a Group 2 nitrate (e.g., $Ca(NO_3)_2$)?

(a) Metal oxide, CO_2, O_2
(b) Metal oxide, NO_2, O_2
(c) Metal hydroxide, N_2, O_2
(d) Metal, NO_2, H_2O

Answer: (b) Metal oxide, NO_2, O_2 — $2Ca(NO_3)_2 \rightarrow 2CaO + 4NO_2 + O_2$.

VII) The reason for the increasing thermal stability of Group 2 nitrates down the group is:

(a) Smaller cations, higher polarizing power
(b) Larger cations, lower polarizing power
(c) Larger size of the nitrate ion (NO_3^-)
(d) Decrease in the charge on the cation

Answer: (b) Larger cations, lower polarizing power — Larger cations distort the nitrate ion less, so more heat is needed to decompose it.

VIII) Which Group 2 sulphate is the least soluble in water?

(a) $MgSO_4$ (b) $BaSO_4$
(c) $CaSO_4$ (d) $BeSO_4$

Answer: (b) $BaSO_4$ — The solubility of sulphates decreases down the group; $BaSO_4$ is insoluble (9.43×10^{-7} mol/100 g).

IX) Which of the following is the least soluble in water?

(a) $Mg(OH)_2$ (b) $Ba(OH)_2$
(c) $MgSO_4$ (d) $BaSO_4$

Answer: (d) $BaSO_4$ — From Table 17.4: $BaSO_4$ 9.43×10^{-7}, $Mg(OH)_2$ 2×10^{-5}, $Ba(OH)_2$ 1.5×10^{-2}, $MgSO_4$ 1.83×10^{-1} mol/100 g of water.

X) Why does Be^{2+} form complexes more easily than Ba^{2+}?

(a) Larger size (b) Lower charge density
(c) Higher charge density (d) More d-orbitals

Answer: (c) Higher charge density — Be^{2+} is the smallest Group 2 ion, so it attracts electron pair donors most strongly.

The heading is printed "MUTIPLE CHOICE QUESTIONS" in the textbook. In MCQ I, barium oxide (the most basic Group 2 oxide) is not among the options; strontium oxide is the correct choice from those given.

Q2. SHORT ANSWER QUESTIONS

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a) Exactly 0.1 moles of calcium hydroxide and barium hydroxide are added to separate beakers containing 100 cm^3 of water and stirred to form saturated solutions. Explain which solution has the higher pH value.

The **barium hydroxide** solution has the higher pH. The solubility of Group 2 hydroxides increases down the group: $Ba(OH)_2$ dissolves up to 1.5×10^{-2} mol per 100 g of water, but $Ca(OH)_2$ only 1.5×10^{-3} mol. Both solutions are saturated (0.1 mol is more than can dissolve), so the $Ba(OH)_2$ solution contains about ten times more dissolved hydroxide. It has a higher concentration of OH^- ions and therefore a higher pH.

b) The size of dichromate ion is larger than the sulphate ion. Following the pattern for the solubilities of Group 2 sulphates, predict which of calcium dichromate and strontium dichromate is more soluble in water?

Calcium dichromate is more soluble. For a large anion such as sulphate, the solubility decreases down the group, because the hydration enthalpy of M^{2+} falls more than the lattice enthalpy. The dichromate ion is even larger, so the same pattern applies: the salt of the smaller cation (Ca^{2+}) is more soluble than that of the larger cation (Sr^{2+}).

c) Write balanced equations for the reaction of Ba with cold water.

$$Ba_{(s)} + 2H_2O_{(l)} \rightarrow Ba(OH)_{2(aq)} + H_{2(g)}$$

d) Why does Be metal form a protective layer on reacting with air while other members do not do so? Explain in terms of the strength of bonding between Be^{2+} and oxide ion.

Be^{2+} is the smallest Group 2 ion (31 pm), so it has the highest charge density. It attracts the O^{2-} ion very strongly, and the bonding in BeO is very strong. The oxide therefore forms a thin, strong layer that holds firmly on the metal surface and stops further reaction. The other M^{2+} ions are larger with lower charge density; their attraction for the oxide ion is weaker, so their oxide layers do not protect the metal and the reaction continues.

The textbook states only that a thin, strong oxide layer protects beryllium; the explanation through charge density follows the textbook's reasoning on ion size.

e) Predict the products of the thermal decomposition of barium nitrate. Write the chemical equation.

The products are barium oxide, nitrogen dioxide (brown gas) and oxygen.

$$2Ba(NO_3)_{2(s)} \rightarrow 2BaO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

f) Why does solubility of Group 2 hydroxides increase in terms of the relative changes in ΔH_{hyd} and $\Delta H_{lattice}$?

Down the group the M^{2+} ion gets larger, so both ΔH_{hyd} and $\Delta H_{lattice}$ decrease. For hydroxides, the decrease in lattice enthalpy is more significant than the decrease in hydration enthalpy. Less energy is needed to break the lattice compared with the energy released on hydration, so the enthalpy of solution becomes more favourable and solubility increases from $Mg(OH)_2$ to $Ba(OH)_2$.

g) In Dow Process, write the balanced equation for the precipitation of magnesium ions from seawater.

$$Mg^{2+}_{(aq)} + Ca(OH)_{2(s)} \rightarrow Mg(OH)_{2(s)} + Ca^{2+}_{(aq)}$$

h) State one medical use of $BaSO_4$ and $Mg(OH)_2$ each.

$BaSO_4$: used in “barium meals” for X-ray imaging of the digestive system; it is opaque to X-rays and safe because it is highly insoluble.

$Mg(OH)_2$: as “Milk of Magnesia” it neutralises excess stomach acid (antacid) and acts as a mild laxative.

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) Why the melting point and density of beryllium are higher than other Group 2 elements?

Beryllium has the smallest atom of the group (metallic radius 112 pm). Its two delocalised valence electrons are closest to the nuclei, so its metallic bonding is the strongest. More heat is needed to break these bonds, which gives the highest melting point (1287 °C). Because the atoms are small and packed closely, more mass is contained in a given volume, so its density (1.85 g/cm^3) is higher than that of magnesium (1.74) and calcium (1.54). The textbook relates these trends to the different packing of the metal atoms down the group.

According to Table 17.1 of the textbook, strontium (2.64 g/cm^3) and barium (3.51 g/cm^3) are denser than beryllium. So the statement about density is true only in comparison with Mg and Ca; the melting point of Be is the highest in the group.

b) Why are Group 1 metals more reactive than Group 2 metals in the same period?

Group 1 metals have only one valence electron (ns^1) and low ionization energies, so they lose it very easily to form M^+ ions. Group 2 metals must lose two electrons (ns^2) to form M^{2+}, which requires more energy because their ionization energies are higher. So Group 2 metals are reactive, but less so than the Group 1 metal of the same period (e.g., Na reacts explosively with water, while Mg reacts well only with steam).

c) Explain why calcium hydroxide is used in agriculture.

Calcium hydroxide, $Ca(OH)_2$ (slaked lime), is a base. It is added to acidic soils to neutralise the acidity and raise the pH, which makes the soil more suitable for crop growth. It also supplies calcium, an essential plant nutrient needed for cell wall development.

$$Ca(OH)_2 + 2H^+ \rightarrow Ca^{2+} + 2H_2O$$

DESCRIPTIVE QUESTIONS

Q4. Describe the reactivity of Group 2 elements with water and dilute hydrochloric acid. Provide relevant chemical equations and differences in their reactivity.

Reaction with water

Except beryllium, Group 2 metals react with water to produce metal hydroxides, $M(OH)_2$, and hydrogen gas.

- **Beryllium** does not react with water or steam, even when heated, because of its protective oxide layer.
- **Magnesium** reacts very slowly with cold water, forming $Mg(OH)_2$ and H_2. With steam it reacts faster, forming MgO and hydrogen.
- **Calcium** reacts with cold water and steam more vigorously than magnesium, forming $Ca(OH)_2$ and H_2. Calcium hydroxide is sparingly soluble and forms a white precipitate.

$$\text{Cold water: } Mg_{(s)} + 2H_2O_{(l)} \rightarrow Mg(OH)_{2(aq)} + H_{2(g)}$$
$$\text{Steam: } Mg_{(s)} + H_2O_{(g)} \rightarrow MgO_{(s)} + H_{2(g)}$$
$$Ca_{(s)} + 2H_2O_{(l)} \rightarrow Ca(OH)_{2(aq)} + H_{2(g)}$$

The hydroxides of Group 2 are less basic than those of Group 1, but their basic character increases down the group due to increasing solubility.

$$Ba_{(s)} + 2H_2O_{(l)} \rightarrow Ba(OH)_{2(aq)} + H_{2(g)}$$

Reaction with dilute hydrochloric acid

Group 2 metals react with dilute acids to produce a salt and hydrogen gas. Reactivity increases down the group.

$$M_{(s)} + 2HCl_{(aq)} \rightarrow MCl_{2(aq)} + H_{2(g)}$$

- **Beryllium** forms $BeCl_2$ and hydrogen, but its protective oxide layer can slow the reaction.
- **Magnesium** reacts readily: $Mg_{(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + H_{2(g)}$; $Mg_{(s)} + H_2SO_{4(aq)} \rightarrow MgSO_{4(aq)} + H_{2(g)}$
- **Calcium** reacts vigorously: $Ca_{(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + H_{2(g)}$
- With sulphuric acid, the reaction of calcium can stop early, because the $CaSO_4$ formed is sparingly soluble and coats the metal with a protective layer: $Ca_{(s)} + H_2SO_{4(aq)} \rightarrow CaSO_{4(s)} + H_{2(g)}$

Differences in reactivity

Metal	With water	With dilute HCl
Be	No reaction with water or steam (protective oxide layer)	Forms $BeCl_2$ and H_2; slowed by the oxide layer
Mg	Very slow with cold water; faster with steam (gives MgO)	Reacts readily
Ca	Reacts with cold water more vigorously; $Ca(OH)_2$ forms a white precipitate	Reacts vigorously
Sr, Ba	React still more vigorously with cold water	React most vigorously

In both reactions the reactivity increases from Be to Ba, because atomic size and shielding increase and ionization energy decreases down the group.

Q5. Explain the thermal decomposition of carbonates and nitrates of Group 2 elements. Give reason and relative order of the ease of thermal stability.

Thermal decomposition is a chemical reaction in which a single compound breaks down into two or more simpler substances when heated. It usually requires energy (endothermic) to break the chemical bonds within the compound.

Carbonates

Group 2 carbonates decompose on heating to form a metal oxide (MO) and carbon dioxide. Their thermal stability **increases down the group**: $BeCO_3$ decomposes at a lower temperature than $MgCO_3$, which decomposes at a lower temperature than $CaCO_3$, and so on. $BaCO_3$ requires the highest temperature.

$$MgCO_{3(s)} \rightarrow MgO_{(s)} + CO_{2(g)}$$
$$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)}$$

Reason: The trend is due to the size of the M^{2+} ion. Smaller, highly charged cations (like Be^{2+}) have a greater **polarizing power**. They distort the large carbonate ion (CO_3^{2-}) to a larger extent, weakening its bonds and making it easier to decompose. As the metal ion gets larger down the group, its polarizing power decreases, so the carbonates become more stable and need more heat.

Metal carbonate	$BeCO_3$	$MgCO_3$	$CaCO_3$	$SrCO_3$	$BaCO_3$
Decomposition temperature (°C) at 1 atm	250	540	900	1289	1360

Nitrates

Group 2 nitrates decompose on heating to form a metal oxide (MO), brown nitrogen dioxide gas (NO_2) and oxygen (O_2).

$$2Ca(NO_3)_{2(s)} \rightarrow 2CaO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

Like the carbonates, the thermal stability of the nitrates **increases down the group**: $Be(NO_3)_2$ is the least stable and $Ba(NO_3)_2$ the most stable. The same polarizing-power argument applies: the smaller M^{2+} ions at the top of the group distort the nitrate ion (NO_3^-) more effectively, so decomposition is easier at lower temperatures.

Order of thermal stability

$$BeCO_3 < MgCO_3 < CaCO_3 < SrCO_3 < BaCO_3$$

$$Be(NO_3)_2 < Mg(NO_3)_2 < Ca(NO_3)_2 < Sr(NO_3)_2 < Ba(NO_3)_2$$

The ease of decomposition is the reverse: beryllium compounds decompose most easily and barium compounds least easily.

Q6. Explain the solubility of sulphates and hydroxides of Group 2 elements, and discuss the trend in their solubility down the group with appropriate examples.

The solubility of an ionic compound in water depends on the balance between two energy changes:

1. **Enthalpy of hydration (ΔH_{hyd}):** the energy released when gaseous ions are surrounded and attracted by water molecules. It is always exothermic. Smaller ions with higher charge density attract water more strongly, so their ΔH_{hyd} is more negative. It becomes less negative down a group as the ions get larger.
2. **Lattice enthalpy ($\Delta H_{lattice}$):** the energy required to break apart one mole of a solid ionic compound into its gaseous ions. It is always endothermic. It depends on the charges and sizes of the ions: when cation and anion are comparable in size the forces are stronger and the lattice energy is higher; when one ion is much larger than the other the forces are weaker.

The **enthalpy of solution (ΔH_{soln})** is the overall energy change when a solid ionic substance dissolves. A more negative or less positive ΔH_{soln} generally favours higher solubility.

Hydroxides: solubility **increases down the group** ($Mg(OH)_2$ is sparingly soluble; $Ba(OH)_2$ is quite soluble). Down the group both ΔH_{hyd} (of M^{2+}) and $\Delta H_{lattice}$ decrease, but the decrease in lattice enthalpy is more significant than the decrease in hydration enthalpy. So less energy is needed to break the lattice, and dissolving becomes more favourable.

Sulphates: solubility **decreases down the group** ($MgSO_4$ is soluble; $BaSO_4$ is insoluble). The sulphate ion (SO_4^{2-}) is relatively large. Down the group both enthalpies decrease, but for a large anion the decrease in hydration enthalpy is more significant than the decrease in lattice enthalpy. So the energy released on hydration falls faster, and dissolving becomes less favourable.

Hydroxide	Solubility at 298 K (mol/100 g water)	Sulphate	Solubility at 298 K (mol/100 g water)
$Mg(OH)_2$	2×10^{-5}	$MgSO_4$	1.83×10^{-1}

Hydroxide	Solubility at 298 K (mol/100 g water)	Sulphate	Solubility at 298 K (mol/100 g water)
$Ca(OH)_2$	1.5×10^{-3}	$CaSO_4$	4.66×10^{-3}
$Sr(OH)_2$	3.4×10^{-3}	$SrSO_4$	7.11×10^{-5}
$Ba(OH)_2$	1.5×10^{-2}	$BaSO_4$	9.43×10^{-7}

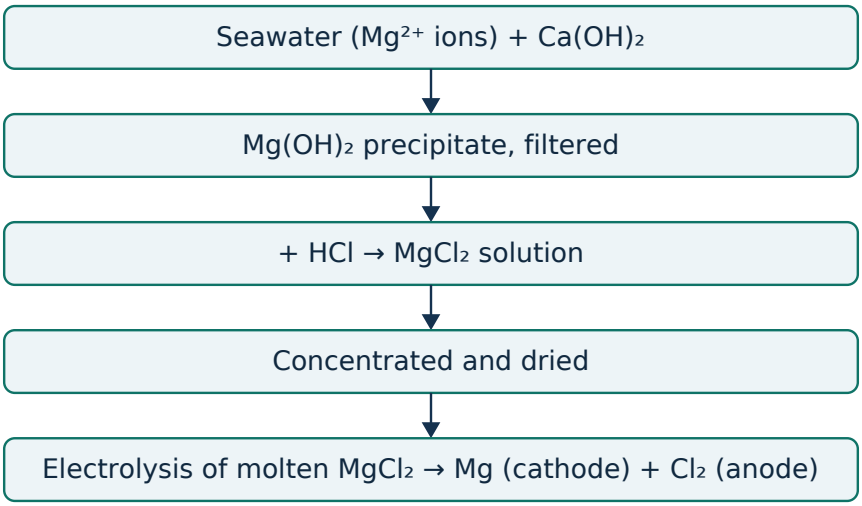
	Hydroxides	Sulphates
Trend down the group	Solubility increases	Solubility decreases
Anion	Small	Large
Dominant change	Lattice enthalpy decreases more	Hydration enthalpy decreases more
Examples	$Mg(OH)_2$ sparingly soluble; $Ba(OH)_2$ quite soluble	$MgSO_4$ soluble; $BaSO_4$ insoluble

Q7. Explain the process of extraction of Mg through Dow's method.

The **Dow process** is a multi-step chemical and electrochemical process that extracts magnesium metal from the vast reserves of Mg^{2+} ions in seawater.

1. **Precipitation:** Seawater is treated with calcium hydroxide, $Ca(OH)_2$ (from limestone or oyster shells), which precipitates magnesium as insoluble magnesium hydroxide.
2. **Conversion to chloride:** The $Mg(OH)_2$ is filtered and reacted with hydrochloric acid to form magnesium chloride.
3. **Concentration and drying:** The $MgCl_2$ solution is concentrated and dried.
4. **Electrolysis:** Molten $MgCl_2$ is electrolysed in large cells. Pure magnesium metal is produced at the cathode and chlorine gas at the anode.

$$Mg^{2+}_{(aq)} + Ca(OH)_{2(s)} \rightarrow Mg(OH)_{2(s)} + Ca^{2+}_{(aq)}$$
$$Mg(OH)_{2(s)} + 2HCl_{(aq)} \rightarrow MgCl_{2(aq)} + 2H_2O_{(l)}$$
$$Mg^{2+}_{(l)} + 2Cl^{-}_{(l)} \rightarrow Mg_{(l)} + Cl_{2(g)} \text{ (electrolysis)}$$



Dow process

The process extracts a valuable metal from an abundant, low-concentration source (seawater) on a massive industrial scale.

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

Quick Check 17.1

a) Write equations for the following reactions: i) Ca with oxygen ii) Ba with water

$$\text{i) } 2Ca_{(s)} + O_{2(g)} \rightarrow 2CaO_{(s)}$$
$$\text{ii) } Ba_{(s)} + 2H_2O_{(l)} \rightarrow Ba(OH)_{2(aq)} + H_{2(g)}$$

b) Predict the extent of the reactions of Sr and Ba with H_2SO_4.

The reactions take place only to a very small extent. The sulphates become less soluble down the group, so $SrSO_4$ and $BaSO_4$ are even less soluble than $CaSO_4$. An insoluble sulphate layer coats the metal almost at once and stops the reaction, although Sr and Ba themselves are very reactive.

c) Explain why Group 2 elements are reactive, based on their electron configuration and ion formation.

They have two electrons in the outermost s-subshell (ns^2). By losing these two electrons they form stable M^{2+} ions. Because these electrons are lost fairly easily, the metals react quickly with other substances.

d Write a balanced equation for calcium and barium burning in air and compare their reactivity.

$$2Ca_{(s)} + O_{2(g)} \rightarrow 2CaO_{(s)}$$
$$2Ba_{(s)} + O_{2(g)} \rightarrow 2BaO_{(s)}$$

Barium is more reactive than calcium: its atom is larger, with more shielding and a lower ionization energy (503 against 590), so it loses its electrons more easily.

e) Predict the density and melting point of the radioactive radium (Ra) following the trend in Group 2.

From Ca to Ba the density rises (1.54, 2.64, 3.51 g/cm^3) and the melting point falls (842, 777, 727 °C). So radium should have a density greater than 3.51 g/cm^3 (about 5 g/cm^3) and a melting point somewhat below 727 °C (about 700 °C).

The accepted values for radium (about 5.5 g/cm^3 and 700 °C) are not in the textbook; they agree with the prediction.

Quick Check 17.2

a) State two physical differences between Group 1 and Group 2 metals.

(i) Group 2 metals are harder; Group 1 metals are very soft and can be cut with a knife. (ii) Group 2 metals have higher densities and higher melting/boiling points than Group 1 metals.

b) State the general formula for a Group 1 oxide and a Group 2 oxide.

Group 1 oxide: M_2O (e.g., Na_2O). Group 2 oxide: MO (e.g., CaO).

c) Write the balanced equation for the thermal decomposition of $SrCO_3$.

$$SrCO_{3(s)} \rightarrow SrO_{(s)} + CO_{2(g)}$$

d) Explain qualitatively why magnesium carbonate is thermally less stable than calcium carbonate.

Mg^{2+} (65 pm) is smaller than Ca^{2+} (99 pm), so it has a greater polarizing power. It distorts the large carbonate ion more, weakening its bonds, so $MgCO_3$ decomposes at a lower temperature (540 °C) than $CaCO_3$ (900 °C).

e) Write the equations for the thermal decomposition of i) magnesium nitrate ii) Barium nitrate. Which of these decomposes at a lower temperature?

$$\text{i) } 2Mg(NO_3)_{2(s)} \rightarrow 2MgO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

$$\text{ii) } 2Ba(NO_3)_{2(s)} \rightarrow 2BaO_{(s)} + 4NO_{2(g)} + O_{2(g)}$$

Magnesium nitrate decomposes at the lower temperature, because the smaller Mg^{2+} ion distorts the nitrate ion more effectively.

Quick Check 17.3

a) Why does $MgSO_4$ dissolve easily as compared to $BaSO_4$?

Mg^{2+} is small with a high charge density, so its hydration enthalpy is highly negative and easily compensates the lattice enthalpy. Ba^{2+} is large, so its hydration enthalpy is much less negative, while the lattice enthalpy (with the large sulphate ion) has decreased only a little. So dissolving $BaSO_4$ is unfavourable and it is insoluble.

b) How does lattice energy affect the solubility of Group 2 sulphates?

Lattice energy is the energy needed to break the solid into gaseous ions; the higher it is, the lower the solubility. In the sulphates the anion is large, so the lattice energy decreases only slightly down the group, while the hydration enthalpy decreases much more. The lattice energy is therefore less and less compensated, and solubility decreases down the group.

c) Which hydroxide is more soluble: $Ca(OH)_2$ or $Sr(OH)_2$? Give reason.

$Sr(OH)_2$ is more soluble (3.4×10^{-3} against 1.5×10^{-3} mol/100 g of water). The solubility of hydroxides increases down the group, because the lattice enthalpy decreases more than the hydration enthalpy.

Quick Check 17.4

a) Write the balanced equation for the electrolysis step in the Dow Process.

$$Mg^{2+}_{(l)} + 2Cl^{-}_{(l)} \rightarrow Mg_{(l)} + Cl_{2(g)}$$

At the cathode: $Mg^{2+} + 2e^- \rightarrow Mg$. At the anode: $2Cl^- \rightarrow Cl_2 + 2e^-$.

b) Write the balanced equation for producing quicklime (CaO) from limestone.

$$CaCO_{3(s)} \rightarrow CaO_{(s)} + CO_{2(g)} \text{ (heat, 900–1000 °C)}$$

c) State one agricultural use of $CaCO_3$ and the chemical principle involved.

$CaCO_3$ is added to acidic soils to increase the pH, making the soil more suitable for crops. The principle is **neutralisation**: the basic carbonate reacts with the acid in the soil.

$$CaCO_3 + 2H^+ \rightarrow Ca^{2+} + H_2O + CO_2$$

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